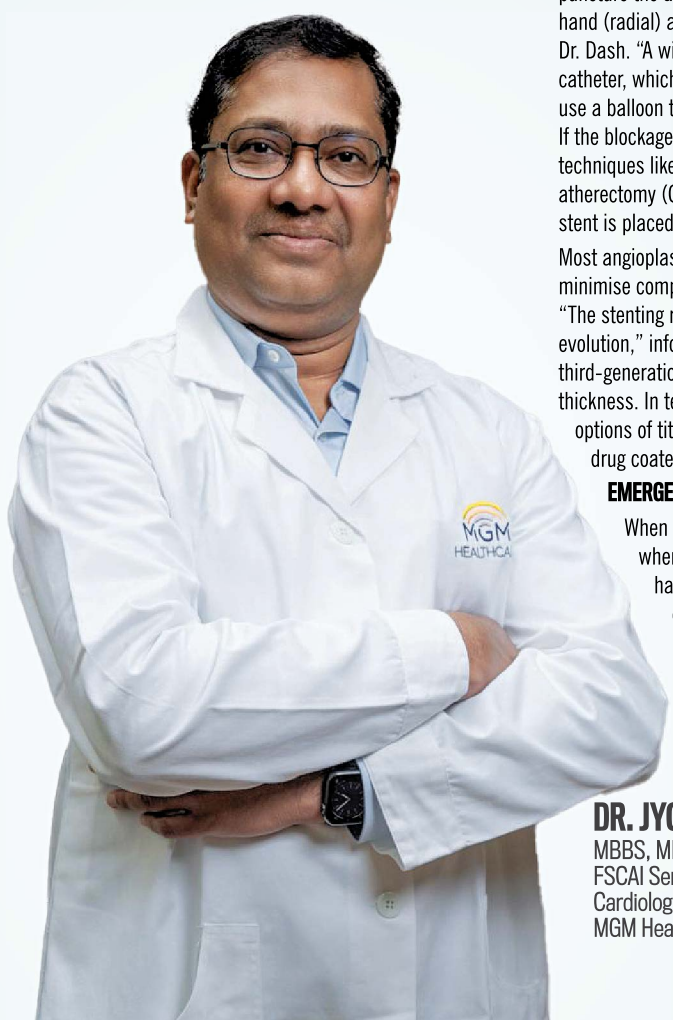
MGM HEALTHCARE

CUTTING-EDGE CARE AT MGM HEALTHCARE'S DEPARTMENT OF CARDIOLOGY

The incidence of heart attack–technically called a myocardial infarction (MI)–has gone up in recent times, even among the young. This is why an awareness of its causes and treatments is essential.

Three major arteries supply blood to the heart: right coronary, left anterior descending and left circumflex. A heart attack occurs when there are blockages in the arteries. A coronary angiogram is done to assess the extent of blockage. After that, the doctor can decide if the condition can be medically managed or requires an angioplasty or bypass surgery. Modern-day angioplasty is low risk, and a preferred choice in many cases.

A multi-speciality hospital like MGM Healthcare – Chennai is well equipped to deal with heart-related emergencies. Vast experience meets new-age technology at its Department of Cardiology, which offers comprehensive cardiac services.



Dr. Jyotirmaya Dash, a renowned Senior Consultant in the Department of Cardiology, MGM Healthcare – Chennai, specialises in Interventional Cardiology with extensive experience spanning over two decades. His expertise encompasses advanced procedures including Primary Angioplasties, Complex PTCA, Chronic Total Occlusion (CTO), Bifurcation Stenting, Left Main Stenting, Structural heart interventions such as TAVI/TAVR, Pacemakers, Endovascular Graft Stenting and Peripheral Interventions.

UNDERSTANDING ANGIOPLASTY

"Like in an angiogram, for an angioplasty we puncture the artery in the leg (femoral) or the hand (radial) and pass a catheter into it," says Dr. Dash. "A wire is then passed into the catheter, which crosses the blockage. We then use a balloon to clear a way in the blocked part. If the blockage is caused by calcification, we use techniques like rotablation or orbital atherectomy (OA). Once the block is cleared, a stent is placed."

Most angioplasties are done via radial (hand) to minimise complications and the hospital stay. "The stenting revolution has outpaced surgical evolution," informs Dr. Dash. "Now we have third-generation stents of 60-65 microns thickness. In terms of materials, we now have options of titanium and cobalt-chromium, all drug coated."

EMERGENCY ANGIOPLASTY

When a patient gets a heart attack where one of the three main arteries has a 100-percent blockage, an emergency angioplasty can be a saviour. When the patient comes to emergency with a STEMI (ST-elevation myocardial infarction)—the other kind of heart attack being NSTEMI (Non-ST-elevation myocardial infarction)—the first order of business is an emergency angiogram. "If we find a 100-percent block, we suck out the clot using an aspiration catheter. If there's still residual blockage, we place an emergency stent and send the patient to CCU for observation and discharge two days later," says Dr. Dash.

"If the patient can be treated within 90 minutes of First Medical Contact (FMC) or what we call 'door-to-balloon' time, then the outcome is excellent. We suggest people call our dedicated emergency number (+91-44-4200-4200) ahead, so we are prepared," says Dr. Dash. An angiogram can be done within 5-10 min and then subsequently an angioplasty.

CHALLENGES AND FUTURE

Sometimes patients have a full blockage in the left anterior descending (LAD) artery or come with a cardiogenic shock. "We take them to the cath lab and give them IABP (intra-aortic balloon pump) therapy. If this doesn't help, we go for ECMO (extra corporeal membrane oxygenation). A third option is the Impella device, a temporary artificial heart pump. These devices help us perform an angioplasty even when pumping function is low," explains Dr. Dash.

On the future of angioplasty, Dr. Dash elaborates, "Slowly we are moving into an era where stents may not be preferred in younger patients. With a CAT RX suctioning machine, we can suck out almost 100 percent of the clot. So, patients who have a clot, but no block can return home without a stent."

The other development is the rise of AI in diagnosis. "By integrating AI, while doing the angiogram we can know the type and severity of the blockage and how it will develop in future," explains Dr. Dash. The Department of Cardiology at MGM Healthcare – Chennai also employs advanced new technologies for angioplasty, including laser (ELCA) therapy and robotics, proving once again that it's the last word in cutting-edge treatments.

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